TIAGO FERRÃO-LINDQVIST

Data Engineering | Test & Validation Engineering

PROFESSIONAL SUMMARY

Test & Validation Engineer with experience building internal Python tooling for engineering test-data processing and reporting, working with SQL Server staging data, and developing Power BI dashboards for validation and design teams. Processes data from six test rigs and supports colleagues through reusable internal tooling. Currently completing an online data engineering course, with six of nine modules finished. Brings a mechanical engineering background in durability and vibration testing, root-cause analysis, pump design and FEA.

TECHNICAL SKILLS

Programming and data processing: Python, pandas, NumPy, SQL, MATLAB, Simulink
Data and reporting: SQL Server, SQLite, Power BI, Matplotlib, npTDMS
Tools: Flask, Docker Compose, Git, Windows Task Scheduler
Engineering: Durability and vibration testing, test-data processing, root-cause analysis, structural FEA

LANGUAGES

Portuguese

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components *2019–Present*

- Conduct durability and vibration testing, support root-cause analysis, and develop tools for processing and reporting test data.
- Built Python scripts to parse TDMS and CSV output from six rigs (about 2 GB per week), flag channels outside test specifications, and generate report figures.
- Set up the scripts to run nightly on a lab PC using Windows Task Scheduler and write summary results to an IT-created staging table in the plant's SQL Server database.
- Packaged the tool for internal use; four other test engineers now use it for their own tests.
- Created a Power BI dashboard using the staging data to show pass/fail trends and cycles to failure by product generation; used by about 15 engineers in validation and design.
- Trained three apprentices in rig safety and test setup (2022–2024).
- Used TIA Portal, with an automation engineer supervising, to adjust PLC setpoints and timers.

Mechanical Design Engineer | Halvik Pumps *2014–2019*

- Designed pump casings and impellers and carried out structural FEA.
- Supported the performance test bench and wrote a LabVIEW data-acquisition program that reduced test setup time from about two hours to 30 minutes.

SELECTED PROJECTS

Rig Data Processing and Reporting Tool | Corvane Automotive Components *2021–Present*

Python scripts process rig output (TDMS and CSV), flag channels outside test specifications, generate report figures, and write summary results to a SQL Server staging table. Runs nightly on a lab PC and is used by four other test engineers.

Technologies: Python, pandas, NumPy, Matplotlib, npTDMS, SQL Server, Windows Task Scheduler

Test Results Dashboard | Corvane Automotive Components *2023*

Built a Power BI dashboard showing pass/fail trends and cycles to failure per product generation, used by about 15 validation and design engineers.

Technologies: Power BI, SQL

Adjuster Drive Load-Profile Model | Corvane Automotive Components *2020–Present*

Developed a MATLAB/Simulink model of a seat-adjuster drive, including motor, gearbox and rail friction, to select rig load profiles matching customer duty cycles.

Technologies: MATLAB, Simulink

Bike-Sharing Trips API | Personal Learning Project *2025*

Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints. Runs locally; followed a tutorial to run it with Docker Compose.

Technologies: Python, Flask, SQLite, Docker Compose, Git

PROFESSIONAL DEVELOPMENT

Online Data Engineering Course | In progress

Completed 6 of 9 modules; no certificate awarded.